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Nationality

UGANDAN

AI Engineer Capability Matrix =

Machine Learning & GenAI

Azure ML, Azure OpenAI, Hugging Face, LangChain, and MLflow for building, deploying, and monitoring AI models.

Data Engineering & Pipelines

Azure Data Factory, Synapse, Databricks, and Delta Lake for large-scale data ingestion and transformation.

MLOps & Automation

CI/CD integration using GitHub Actions & Azure DevOps, containerised deployment via AKS & Docker, model versioning with MLflow.

Cloud Infrastructure:

Designing secure, scalable AI environments with VNETs, Managed Identities, Private Endpoints, and RBAC in Azure

Observability & Governance

Model monitoring, drift detection, and compliance alignment (FCA, ISO 27001) using OpenTelemetry and Prometheus

Programming & Frameworks

Python, PySpark, TensorFlow, PyTorch, Scikit-learn, REST/GraphQL APIs for integration and automation.

AI Solution Design:

Translating business needs into intelligent automation and decision systems that deliver measurable outcomes.

DevOps & Collaboration

Azure DevOps and GitHub for CI/CD of data pipelines and notebooks, Terraform and Bicep for Infrastructure as Code, Jira or Azure Boards for Agile delivery and sprint planning.

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AI Platforms & MLOps | Data Engineering & Automation | Generative AI & LLM Systems | Cloud-Native Solutions (Azure | AWS) | Finance & FinTech | Public Sector | Aerospace & Defence | Digital Transformation Enablement |

AI Engineer and Solutions Architect helping organisations use artificial intelligence to work smarter, faster, and more securely. I design and deliver AI-powered systems that automate processes, reduce risk, and turn data into meaningful insights for better decisions. I work closely with business leaders and technical teams to ensure every solution supports real goals, improving efficiency, compliance, and customer experience.

Experience

OCT 2024 - Present

AI Solutions Engineer | SymphonyAI

Contributing to the design and delivery of enterprise AI solutions that help global banks and investment firms strengthen fraud detection, risk analysis, and compliance automation within the SymphonyAI Financial Services division.

Key Contributions at SymphonyAI (Financial Services vertical):

- Architecting end-to-end AI workflows on Azure AI and MLflow to unify fragmented model pipelines across global financial clients addressing the challenge of manual deployment and inconsistent governance by standardising how models are trained, validated, and released.
- Designing retrieval-augmented generation (RAG) solutions using LangChain and Azure Cognitive Search to tackle the opacity of AI decisioning enabling auditors and compliance officers to query AI reasoning and trace the source of each insight in natural language.
- Developing autonomous AI agents with Python orchestration to automate high-volume compliance checks, sanctions screening, and risk reporting reducing turnaround times from days to minutes while maintaining full auditability.
- Embedding observability, lineage, and governance controls through OpenTelemetry, Prometheus, and Azure Monitor to overcome a lack of visibility across AI pipelines ensuring every model in production could be traced, explained, and defended to regulators.
- Collaborating with Chief Data Officers, risk leads, and regulators to co-design responsible AI frameworks aligned with FCA, PRA, and ISO 27001 standards bridging the gap between technical engineering and enterprise compliance requirements.

Impact:

Delivering a secure, transparent, and scalable AI foundation for financial institutions reducing regulatory risk, accelerating compliance reporting, and improving the speed and accuracy of fraud and risk decisioning across the enterprise

JLY 2023 - OCT 2024

AI & Cloud Solutions Engineer | Airbus Defence And Space

Designing and delivering sovereign AI solutions across defence and satellite operations, using Microsoft Azure AI Foundry, Copilot Studio, and OpenAI Services to enable secure, mission-ready intelligence systems within multi-domain, classified environments.

Key Contributions to Airbus Defence and Space:

- Designing and developing AI architectures on Azure AI Foundry and Copilot Studio to address fragmented legacy workflows providing mission teams with secure, cloud-native platforms for AI-assisted analytics and decision-making.
- Building retrieval-augmented generation (RAG) pipelines using Azure OpenAI, Cognitive Search, and vector databases to overcome slow, manual data retrieval enabling operators to query mission data and satellite telemetry in natural language with explainable outputs.
- Developing automation scripts and microservices in Python to replace repetitive operational tasks improving efficiency and reducing human error across mission support systems.
- Implementing full AI lifecycle management with MLOps and CI/CD practices to solve deployment bottlenecks ensuring all AI models are versioned, monitored, and continuously improved within sovereign data boundaries.
- Integrating AI observability and governance frameworks through Azure Monitor, OpenTelemetry, and Prometheus to address transparency and accountability gaps allowing command teams to track performance, bias, and reliability of deployed AI systems.
- Collaborating with cybersecurity, flight operations, and data engineering teams to align AI solutions with NATO and Airbus sovereign standards ensuring compliance, resilience, and operational readiness across multi-domain defence networks.

Impact:

Delivered sovereign AI systems that enhanced mission resilience and interoperability, allowing Airbus to integrate AI into classified command environments without compromising national security.

Certifications

- Azure AI Engineer Associate
- Azure Data Engineer Associate
- Azure Administrator Associate
- CKA (Certified Kubernetes Administrator)
- Docker Certified Associate
- HashiCorp Terraform Associate

Enterprise Trained

- Azure Machine Learning & MLOps
- Azure OpenAI & Cognitive Services
- Azure Security Engineer
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Agile & Scrum Ways of Working

- **Collaborative Delivery**
Work closely with data, cloud, and product teams to design and deliver AI solutions that directly support business goals such as automation, insight generation, and efficiency improvements.
- **Sprint Planning & Prioritisation**
Break large AI initiatives into smaller, achievable sprint goals, ensuring each cycle delivers visible progress and measurable business outcomes.
- **Stakeholder Engagement**
Partner with business leaders, analysts, and product owners to understand challenges, set priorities, and translate technical AI capabilities into clear business value.
- **Transparent Progress & Feedback**
Contribute to stand-ups, reviews, and retrospectives to keep everyone aligned, share updates in plain language, and adapt plans quickly based on feedback.
- **Quality & Continuous Improvement**
Build reliability and trust into AI systems by automating quality checks, monitoring results, and continuously refining models and data processes as the business evolves.

APR 2022 - JUL 2023

AI Platform Engineer | DWP Digital

DWPDigital underpins the UK's largest citizen-facing platforms, including Universal Credit, Pensions, and Benefits. I helped design AI and automation solutions to improve fraud detection, service delivery, and transparency within welfare systems.

Key Contributions to DWP Digital Services:

- Developed and deployed real-time AI pipelines using Kafka and Spark, enabling proactive fraud detection across welfare and pension claim systems.
- Built AI model orchestration frameworks on Azure Machine Learning and MLflow, automating training, deployment, and retraining across multiple citizen-facing services.
- Created explainability dashboards integrating Grafana and OpenAI APIs, helping policy leaders visualise and justify AI-driven decisions in compliance with GDS standards.
- Automated infrastructure provisioning with Terraform and GitHub Actions, improving release reliability for AI models without service downtime.
- Worked cross-functionally with policy, risk, and technology teams to align data ethics and algorithmic fairness with DWP's public-service mission.
- Documented and modelled AI ecosystems in Archimate, mapping interactions between data, services, and policy to support long-term AI strategy.

Impact: Improved accuracy and fairness of welfare processing, reduced fraud losses, and enhanced public trust in DWP's use of AI for citizen services.



MAY 2020 - APR 2022

AI Systems Engineer | Monzo Bank

At Monzo, one of the UK's leading digital banks, I contributed to the development of AI systems that improved fraud detection, compliance automation, and customer experience through data-driven intelligence.

Key Contributions to Monzo Digital Bank:

- Engineered real-time fraud detection pipelines using Kafka, Flink, and PyTorch, processing millions of transactions daily to identify anomalies instantly.
- Implemented NLP-powered support automation using Hugging Face Transformers, cutting manual query resolution time for operations teams.
- Built explainability tooling and model-monitoring dashboards for FCA audits, providing visibility into model logic and bias detection.
- Integrated MLOps practices through GitHub Actions and MLflow, ensuring reproducibility and traceability for all production models.
- Collaborated with risk and compliance teams to align AI pipeline design with financial regulatory frameworks.
- Contributed to data-driven product innovation, enabling Monzo to roll out new personalised financial features powered by behavioural AI.

Impact: Strengthened Monzo's fraud prevention and regulatory posture while enabling faster, data-informed innovation in customer-facing financial products.



SEPT 2018 - MAY 2020

AI&CloudPlatformEngineer | The Game Company

TGC is an AI-powered cloud gaming platform. I designed and automated systems that used data and machine learning to enhance gameplay experience, matchmaking accuracy, and player retention.

Key Contributions to The Game Company (TGC):

- Built low-latency AI recommendation systems with TensorFlow and Redis Streams, optimising matchmaking and content delivery globally.
- Architected real-time analytics pipelines for player engagement telemetry, providing game designers with actionable insight.
- Deployed AI-driven platform updates through Kubernetes, ArgoCD, and AWS/GCP pipelines, achieving sub-minute global rollouts.
- Developed reinforcement learning agents to dynamically adjust game difficulty and player experience.
- Worked with data scientists and developers to productionise AI prototypes into scalable, monitored services.
- Improved cross-region performance through CDN and caching strategies integrated with AI load balancing.

Impact: Enhanced player retention and experience by integrating adaptive AI systems into live operations — directly improving engagement metrics and revenue performance



May 2017 - Sept 2018

Automation Engineer | ABB (Industrial Automation)



At ABB, I worked on industrial automation projects that combined IoT, AI, and predictive analytics to optimise energy systems and production reliability across manufacturing and utilities clients.

Key Contributions to ABB Digital Services:

- Developed IoT data pipelines using Azure IoT Hub and Databricks, connecting thousands of field devices for live telemetry.
- Applied AI-driven predictive maintenance models with TensorFlow and Scikit-learn, reducing unplanned downtime for energy clients.
- Automated training and deployment pipelines through Azure DevOps and Terraform, standardising MLOps across industrial plants.
- Built digital twins of industrial systems, enabling simulation-based diagnostics and performance optimisation.
- Collaborated with operations teams to translate maintenance knowledge into AI rule engines and alerts.
- Implemented edge-to-cloud data security standards, ensuring reliable, compliant transmission of sensitive operational data.

Impact: Improved system reliability and reduced downtime for clients through predictive AI solutions that optimised operations across energy and manufacturing networks.

JULY 2014 - APR 2017

Cloud Engineer | Cargill



One of the world's largest food and agriculture companies, operating across 70+ countries. As a Cloud Engineer at Cargill, I supported the design and automation of AWS based platforms powering food supply, processing, and sustainability initiatives. Focussing on cloud migration, performance optimisation, and security, enabling reliable digital solutions across Cargill's global operations.

Key Contributions to Cargill Supply Chain:

- Supported the migration of supply chain applications to AWS, reducing infrastructure costs by 22% and improving scalability for global logistics and food processing systems.
- Automated infrastructure builds with Terraform cutting provisioning time by 55% and enabling faster delivery of digital solutions to agriculture and sustainability teams.
- Improved system performance and reliability by optimising autoscaling groups, storage tiers, and load balancers, reducing application downtime by 35% and enhancing efficiency in food production and distribution workflows.
- Implemented cloud monitoring and observability using AWS CloudWatch, Azure Monitor, and custom dashboards, reducing mean time to detection (MTTD) and recovery (MTTR) by 30%, giving global operations teams real-time visibility.
- Strengthened cloud security compliance by deploying IAM controls, encryption policies, and automated vulnerability scans, reducing critical security findings by 28% and ensuring alignment with regulatory and corporate standards.

"AI Engineer experienced in designing and deploying intelligent, data-driven solutions that combine machine learning, automation, and cloud technologies. Proven track record of improving system performance, scalability, and decision accuracy."